

Predictive Analytics for Employee Attrition: A Comparative Study of Machine Learning Algorithms

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Abstract- Employee attrition is a significant and widespread challenge that organizations face globally. This issue significantly affects productivity, team dynamics, and overall profitability. The main objective of this research project is to tackle the significant issue of employee attrition. This undertaking involves conducting a thorough analysis using a dataset from a fictional organization specifically designed to mimic real-world HR situations closely. The research employs predictive analytics and machine learning techniques to identify possible cases of attrition. It also provides managerial recommendations to improve employee retention strategies.

This study evaluates the efficacy of different machine learning algorithms, specifically Random Forest, K-Nearest Neighbours (K-NN), and Naive Bayes, in forecasting employee attrition. Moreover, the methodology being proposed integrates the utilisation of an ensemble technique. This approach integrates the favourable attributes of multiple models, leading to enhanced predictive accuracy. Based on the research outcomes, the Naive Bayes model exhibits a higher level of effectiveness than alternative models, particularly when employing meticulously chosen features. The model as mentioned earlier demonstrates a significant level of precision in its ability to forecast attrition.

The study suggests that it would be beneficial to investigate advanced prediction models, employee sentiment analysis based on natural language processing, and personalised staff engagement initiatives. To monitor attrition patterns over time, it is necessary to conduct longitudinal attrition studies. This tool assists companies in developing effective retention strategies that can adapt to evolving circumstances. Implementing these measures can effectively reduce employee turnover and contribute to developing a stable and thriving team.

Keywords- Employee attrition, Predictive analytics, Machine learning, Naive Bayes (NB), Random Forest(RF), K-Nearest Neighbors (K-NN), Employee sentiment analysis, Feature engineering, Ensemble models, Feature selection

I. INTRODUCTION

The issue of employee attrition is a matter of great worry for businesses on a global scale, as it can result in substantial ramifications for productivity, team cohesion, and overall organizational effectiveness. The departure of people from an organisation results in losing key skills and expertise, necessitating the recruitment and training of replacements, which incurs expenses and delays. The collection of extensive data by Human Resource departments poses a significant problem in effectively identifying appropriate candidates for replacement.

This research article investigates the issue using diverse machine learning models to forecast the likelihood of an employee's departure from the organization. By utilizing these prognostications, organizations can formulate focused retention plans for diverse personnel, thereby augmenting their capacity to address attrition more efficiently. The significance of this study rests in its ability to equip organizations with valuable tools and managerial insights to address attrition effectively.[8]

The present paper is organised as follows: Section I presents the introduction. Section II provides an overview of the literature that is relevant to this study. Section III outlines the machine learning models that were used in this study. Section IV presents and analyses the results that were obtained. Section V explores the implications of the findings for management. Lastly, Section VI offers a concluding summary of the article. By leveraging predictive models and data-driven initiatives, organisations can create a positive work environment, enhance employee engagement, and ultimately improve staff retention.

II. LITERATURE REVIEW

In the contemporary era of information-based industries, the workforce is widely acknowledged as a crucial resource,

and the departure of personnel can influence an organization's competitive edge. Researchers have devised multiple frameworks and predictive models to comprehend the underlying causes of attrition, assess its financial implications, and recommend viable measures for employee retention.[12]

To forecast employee attrition, data mining methodologies are employed to analyze the identical dataset, resulting in disparate outcomes. In a particular study, an examination was conducted to compare approaches, including Naïve Bayes, Support Vector Machines, Logistic Regression, Decision Trees, and Random Forests. The findings of this investigation indicated that Support Vector Machines were deemed as the preferred methodology for the unique dataset employed.[13]

A study used logistic regression techniques to analyze demographic data of both separated and existing employees. The researchers formulated a risk equation to forecast the likelihood of employee attrition, which was subsequently employed to assess the attrition risk within the existing workforce. Through the identification of clusters that exhibit high risk, the researchers were able to develop strategic action plans aimed at mitigating attrition.[4]

A theoretical framework to forecast employee attrition is proposed by identifying statistically significant variables associated with an employee's choice to voluntarily terminate their employment. The researchers utilized machine learning algorithms such as Principal Component Analysis, K-Nearest Neighbours, and Random Forest. The Logistic Regression model demonstrated its best level of accuracy, reaching a success rate of 74%.[1]

In the current research study, machine learning techniques are being utilized to predict employee turnover. The primary objective is the development of a model aimed at predicting employee turnover rates. Random Forest and AdaBoost classifiers are currently being employed to achieve precise prediction outcomes. This research paper sheds light on the factors that exert influence on employee attrition within organizations, thereby offering valuable insights for top management to make well-informed decisions concerning the retention of their workforce.[10] One of the research studies delves into the significance of mitigating employee attrition in corporate settings through the application of machine learning algorithms, with a particular focus on Random Forest. It makes use of HR analytics data from IBM Analytics, encompassing 35 distinct features and a sample pool of 1470. To enhance model equilibrium, the research applies the SMOTE mechanism, leading to noticeable enhancements in training metrics.[11]

Furthermore, a particular study was conducted to examine employee turnover at call centers, which are widely recognized for their elevated turnover rates. The researchers examined the operational performance records of sales agents who observed that the Naïve Bayes and Random Forest classifiers exhibited comparable performance in predicting turnover. The efficacy of the Naïve Bayes

algorithm was demonstrated to be competitive when additional information was incorporated despite its inherent simplicity.[3]

III. METHODOLOGY

A. Experiment Tools

This study uses Jupyter Notebook to conduct systematic and controlled experiments to forecast employee turnover using machine learning approaches. The present research work integrates a variety of essential libraries, including numpy, pandas, seaborn, matplotlib.pyplot, LabelEncoder, train_test_split, accuracy_score, confusion_matrix, RandomForestClassifier, GridSearchCV, and StandardScaler to streamline the examination and manipulation of data.

B. Stages for Attrition Analysis

The method of analysis is comprised of six sequential steps, which are outlined as follows:

- **Data Collection:** The initial step involves the gathering and examination of data to gain a better understanding of the study's primary objectives.
- **Data Pre-processing:** This process involves various tasks, including cleaning the data, data transformation, and the generation of new variables that can offer insightful information for subsequent processes of analysis.
- **Exploratory Data Analysis (EDA):** During this phase, the dataset is analyzed to provide textual and visual summaries that highlight significant characteristics of the data.
- **Feature Engineering:** In this stage, pertinent and significant attributes are carefully constructed to improve the data's quality for the purpose of analysis.
- **Model Building:** This stage encompasses the development and improvement of machine learning models.
- **Model Evaluation:** The efficacy of the presented models is evaluated through performance assessment.

IV. DATA ANALYSIS

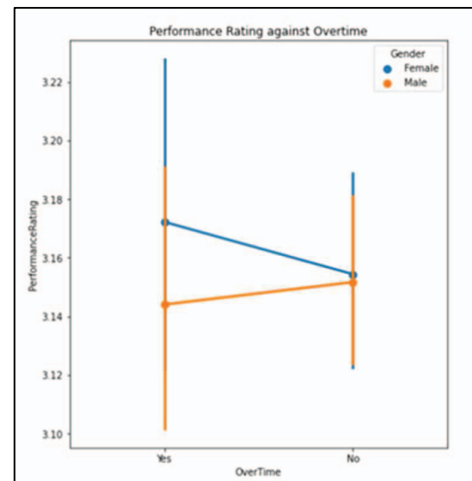
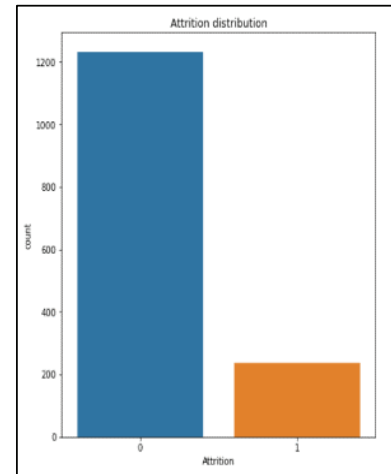
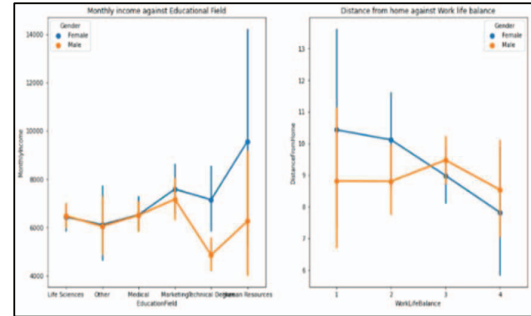
A. Attrition Analysis Steps

- **Data collection:** The process of data collection encompasses the acquisition of data from diverse sources. The data utilized in this study was sourced from Kaggle, specifically, a dataset provided by IBM. The dataset encompassed a total of 1470 rows and 35 columns. Although the dataset is fictional, it is designed to simulate real-world human resources (HR) issues. The attributes utilized in this dataset are both commonplace and pertinent to HR departments across various organizations.
- **Exploratory Data Analysis:** Exploratory data analysis refers to the systematic examination of data in order to identify and understand patterns, trends, and

relationships that may be present within the data. The primary objective is to gain significant insights from the unprocessed data. The present study involved a series of crucial procedures in the data analysis process.

- 1) **Data Preprocessing:** The initial stage of data preprocessing encompassed the essential tasks of data cleaning, filtering, and transforming, which were undertaken to ensure the data was suitably prepared for subsequent analysis. The execution of tasks such as managing missing values, scaling variables, and transforming data into a compatible format was undertaken. The dataset was imported using the "pd.read_csv" function, and the presence of missing values was assessed using the "dataset.isnull().any()" method. Missing values were addressed by the utilization of techniques such as "dataset.isnull().sum()" and "dataset.dropna()"; nonetheless, it is worth noting that no missing values were identified within the dataset.
- 2) **Data Type Conversion:** The process of converting categorical data into numerical form was undertaken in order to render it compatible with machine learning models. The input features, denoted as "X," were partitioned or distinguished from the output variable, denoted as "y," in order to facilitate the model's comprehension of the features and its ability to make predictions. The OneHotEncoder method was selected to transform categorical input into a numerical representation, hence mitigating the risk of erroneous assumptions made by the models. The procedure was implemented with the pandas library.
- 3) **Data Visualization:** The utilization of data visualization is of paramount importance in the field of machine learning, as it facilitates the comprehension and interpretation of data patterns, correlations, and trends through visual means(fig.1). By conducting a comprehensive study of the data, a number of significant observations can be discerned:
 - a) The variable MaritalStatus exhibits comparable consequences to Age, particularly in relation to promotions and potentially other facets of the dataset. It is advisable to contemplate the elimination of one of these variables to mitigate duplication and multicollinearity inside the model.
 - b) The present study examines the potential influence of overtime on male relationship satisfaction and work-life balance. To enhance the efficiency of the model and mitigate any concerns related to multicollinearity, it may be prudent to contemplate the exclusion of one of these variables.
 - c) There is a notable association between the field of education pursued and the monthly income earned. Given the strong correlation between these variables, it may be superfluous to incorporate both into the study.
 - d) The impact of overtime work is observed to differ across several variables and demonstrates significant disparities across individuals of different genders. Thorough deliberation is important when analysing the ramifications of overtime on several facets of employee conduct and welfare.

- e) The attrition distribution in the workplace indicates that a mere 16% of employees have seen attrition. This observation suggests that there is a significant imbalance in the distribution of the output variable (attrition), which has the potential to impact the model's predictive accuracy for attrition.



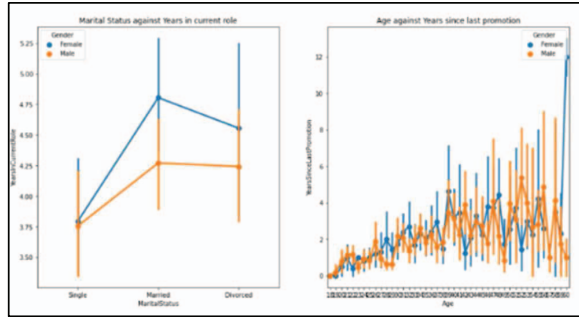


Fig. 1. – Data Visualization Result

- **Feature engineering:** Feature engineering plays a crucial role in the process of data preparation within the field of machine learning. The process entails the conversion and depiction of data in an optimal manner with the aim of enhancing the performance of a model. The aforementioned procedure integrates domain expertise, intuition, prior knowledge, and logical reasoning to generate pertinent and significant characteristics that have the potential to augment the predictive capability of the model.
- **Feature selection:** The process of selecting features is a crucial phase in constructing machine learning models. The process entails selecting the most pertinent features from the dataset to construct an ideal model. Several methodologies, such as implementing the random forest model, can be employed to ascertain the relative significance of individual features. The process of selecting optimal features in a model guarantees that attention is placed on the factors that have the most influence, potentially enhancing accuracy and efficiency.
- **Model Building:** During the model building process, a classification model is constructed. Subsequently, the constructed machine learning model is employed on the dataset to generate predictions and acquire outcomes. Overfitting is widely recognised as a significant challenge in the field of machine learning. There is a necessity to develop a generalized model that can yield favorable outcomes on the test data as well. The utilization of random forest methodology addresses this predicament by amalgamating many decision trees, resulting in reduced bias and variation.

V. RESULTS

Random Forest

The Random Forest algorithm is an ensemble learning technique that effectively mixes many decision trees to get predictions that are both more accurate and stable. The utilization of this approach mitigates the potential for overfitting and demonstrates adaptability and extensive application, even in the absence of hyper-parameter adjustment [9]. The construction of the random forest classifier was accomplished with success, hence facilitating the forecasting of employee turnover for datasets of unknown kinds. The model attained a training accuracy of roughly 86%. In order to effectively assess the model's effectiveness, it is imperative to take into account the number of attrition cases it accurately anticipated. The

utilization of the confusion matrix proves to be a highly effective method for achieving this objective. The main diagonal of the matrix corresponds to accurate forecasts, including both true positives and true negatives. On the other hand, the items outside the main diagonal indicate inaccurate predictions, encompassing both false positives and false negatives. The confusion matrix offers a thorough perspective on the performance of the model and aids in evaluating its efficacy in accurately predicting attrition.

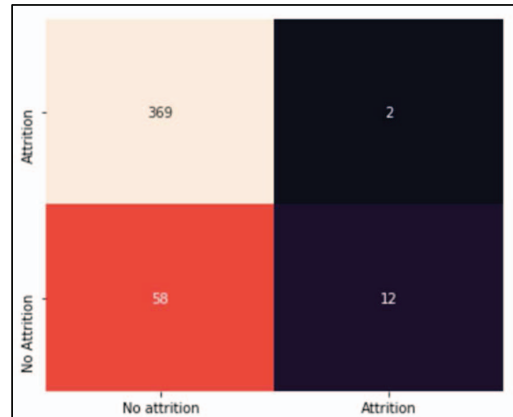


Fig. 2- Prediction result for random forest classifier

The analysis of the results indicates the following:

Among the 381 instances denoting "No Attrition," the model accurately categorized 369 instances as "No Attrition" while erroneously categorizing 12 instances as "Attrition."

Among the 60 entries that encompass real instances of "Attrition," the model accurately identified 2 items as "Attrition," while erroneously categorizing 58 entries as "No Attrition."

The primary aim of the approach is to discern situations in which staff attrition is prone to transpire. The process of assessing the significance of features aids in comprehending the extent to which variables impact predictions. The model possesses an inherent capability to assess the significance of individual features.

The iterative and dynamic nature of refining feature selection and model evaluation is evident in the process. The research involves conducting many iterations of training and testing numerous models with different combinations of characteristics. The objective is to identify the most influential features and develop a robust model that effectively predicts employee attrition. The use of continuous analysis and adjustments has the potential to greatly improve the prediction capabilities of the model.

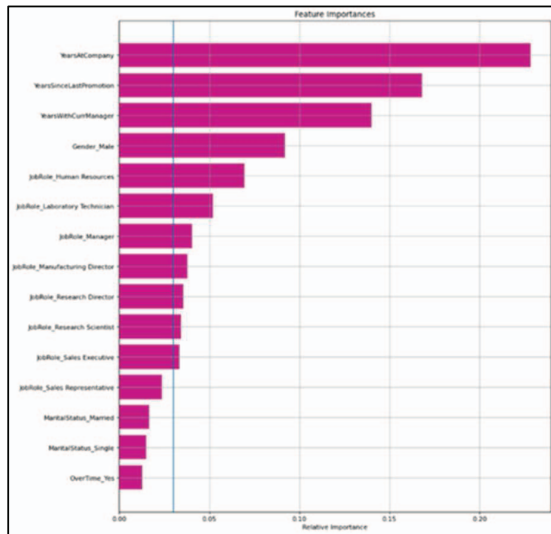


Fig. 3- Evaluating the weight of the features.

The graph illustrates a positive correlation between higher relative importance and improved performance. The data is subjected to manipulation in order to generate a novel feature map that exclusively includes pertinent inputs. The narrative implies that individuals who maintain a static role for prolonged durations, as shown by their tenure within the organization, time elapsed since their previous promotion, and duration of their present managerial relationship, are more prone to encountering challenges. This discovery is consistent with the widely accepted notion that individuals tend to experience a decrease in satisfaction when engaging in repetitive activities for extended periods.

The process of feature scaling is a technique commonly used in data preprocessing to normalise the range of features or variables. It involves transforming the human resources datasets, it is common for features to exhibit varying sizes, a characteristic that has the potential to impact the performance of models. The presence of substantial differences in size across features has the potential to impede the effectiveness of optimisation procedures, such as gradient descent, and adversely affect the performance of some models when applied to unnormalized data. In order to address this concern, a proportionally adjusted version of the feature map is produced, thereby guaranteeing that all features possess comparable scales. The normalization technique improves the model's performance by enabling more efficient training and mitigating potential biases arising from varying scales.

Model Evaluation

The selected statistic for this research is the F1 score, a composite measure combining precision and recall through a weighted average. The measure of recall assesses the comprehensiveness of a model by quantifying the ratio of correctly predicted positive instances to the total number of actual positive instances in the test dataset. Multiple grid search models are constructed, and the performance of these models is assessed using confusion matrix plots.

Furthermore, the K-Nearest Neighbours (K-NN) and Naive Bayes approaches are also delineated.

The random forest classifier has been constructed and evaluated using both the complete collection of features and a new set of features. The outcomes are now evaluated by grid search in order to optimize the hyperparameters, which encompass `n_estimators`, `max_features`, `max_depth`, `min_samples_split`, `min_samples_leaf`, and `bootstrap`.

Upon doing a grid search on the random forest model utilizing all available characteristics, the resulting F1 Score value is deemed unsatisfactory. Following that, the identical grid search technique is employed on the random forest model that has been constructed using a specific set of features[5]. The precision achieved with the chosen set of features is marginally lower compared to the prior model, but, it exhibits a superior F1 Score (table III). However, out of all the RF models, the "Random Forest CM (Selected Features)" seems to be the most optimal choice.

K-NN

The K-Nearest Neighbours (K-NN) algorithm is sometimes referred to as a lazy learner because of its characteristic of not constructing any explicit models. Instead, it relies on identifying the nearest instances from the training dataset to make predictions for new instances, based on the characteristics of these neighboring instances [7]. The K-Nearest Neighbours (K-NN) technique is a non-parametric method that does not rely on assumptions on the distribution of the underlying data. The algorithm clusters elements by considering their K nearest neighbors, employing a customizable distance metric such as Euclidean distance. In the context of K-Nearest Neighbours (K-NN), it is customary to include scaled features, necessitating the normalization of data prior to model training.

The initial K-Nearest Neighbours (K-NN) model is constructed by utilizing all the available features, with a value of K set to 5. The resulting accuracy values are presented in the accompanying table. Upon careful analysis of the confusion matrix, it becomes evident that superior outcomes are attained when employing the chosen set of features.

Following this, two basic K-Nearest Neighbours (K-NN) models are constructed, incorporating grid search methodology to optimize the hyperparameters, including `n_neighbors`, `weights`, `algorithm`, `metrics`, and `p`. The outcomes derived from the grid search are presented in table II. The grid search is conducted on a K-NN model that is constructed using the chosen set of features. The outcomes of this process are presented in the table. Despite observing some improvement in performance using the chosen set of characteristics, the F1 score still exhibits a comparatively low value of 41%. In order to gain a deeper comprehension of the model's performance, an examination and evaluation of the confusion matrix is undertaken.

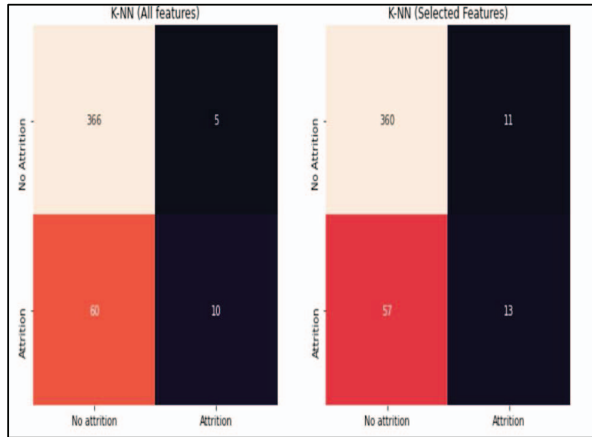


Fig. 4- Prediction result for KNN model

Naive Bayes

The Naive Bayes algorithm is widely used in classification tasks, as it leverages the probabilities of events to categorize instances. The model has strong performance when dealing with intricate datasets that pose difficulties for conventional learning algorithms [2].

The Naive Bayes algorithm is a collection of algorithms that operate on the assumption of feature independence. The Naive Bayes classifier assumes that each feature makes an independent contribution to the prediction without considering any potential relationships between features.

The Naive Bayes model is characterized by its simplicity, and it does not exhibit substantial improvements when subjected to hyperparameter adjustment by grid search. Nevertheless, it is capable of yielding rather satisfactory outcomes. A comprehensive model incorporating all relevant features was constructed with an accuracy of 75.056%. The findings are presented in a tabular format.

The Naive Bayes model demonstrates satisfactory accuracy in the context of this research endeavor, given that the features exhibit no statistically significant interdependence. This condition is a fundamental assumption for the model to yield favorable performance.

Subsequently, the evaluation of the model's performance is conducted only based on the chosen features. The utilization of a confusion matrix offers additional elucidation into the performance of the model, as it provides information on the accuracy of the model's classification of both "attrition" and "no attrition" instances.

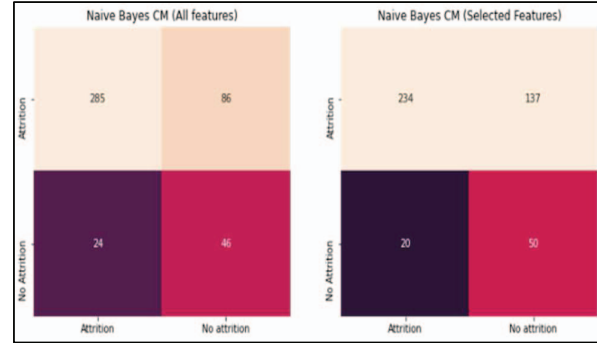


Fig. 5- Prediction result for Naive Bayes model

Although the Naive Bayes model did not exhibit outstanding accuracy for the given dataset, it demonstrated a rather high precision accuracy of 76% in forecasting attrition when utilizing all the characteristics. This precision accuracy closely corresponds to the overall accuracy of the model, which stands at 75%.

Nonetheless, when the model was executed along with the chosen characteristics, it yielded an improved attrition forecast accuracy of 83%. However, the overall prediction accuracy experienced a notable decline, rendering it less formidable as an independent model.

In order to augment the predictive capability, a bespoke ensemble methodology is employed, which amalgamates the advantageous attributes of many models. Ensembling refers to the practice of integrating many models in order to enhance the overall performance. [14] They have the potential to mitigate both variation and bias.[6] Two different ensemble combinations are utilized in the experiment.

TABLE I. Building an Ensemble model

Ensemble1	Ensemble2
- Naive Bayes (all features) with a weight of 0.5 - K-NN (grid search + all features) with a weight of 0.1 - Random Forest (grid search + all features) with a weight of 0.3	- Naive Bayes (selected features) with a weight of 0.5 - K-NN (selected features) with a weight of 0.2 - Random Forest (grid search + selected features) with a weight of 0.3

The Naive Bayes algorithm was selected as the foundational classifier with the greatest weight, primarily because of its notable accuracy in forecasting attrition. The primary aim is to develop a model that exhibits a substantial level of accuracy across all metrics, while also demonstrating efficacy in forecasting attrition.

The initial ensemble, constructed using the chosen features, yielded a classification accuracy of 80.95%. In contrast, the subsequent ensemble, employing models with a specific set

of features, achieved a higher accuracy rate of 83.67%. The utilization of ensemble models has the potential to enhance overall accuracy while effectively preserving attrition forecast skills (table I & II)

TABLE II. Ensemble model accuracy prediction

Ensemble	Weights	Ensemble's Accuracy
NB+KNN+RF	0.5+0.1+0.3	80.95238095238095%
NB+KNN+RF	0.5+0.2+0.3	83.6734693877551%

In order to evaluate the effectiveness of these ensemble models, an examination of the confusion matrices is conducted. The efficiency of ensemble models in predicting employee attrition can be determined through the use of confusion matrices. This enables the comparison between the observed and forecasted values, facilitating the evaluation of the model's capabilities and limitations. The results indicate that two models with comparable characteristics were generated, both demonstrating satisfactory levels of accuracy (81% and 84% respectively). These models were able to achieve a minimum accuracy rate of 50% in forecasting attrition, surpassing all other models with the exception of Naive Bayes (table III)

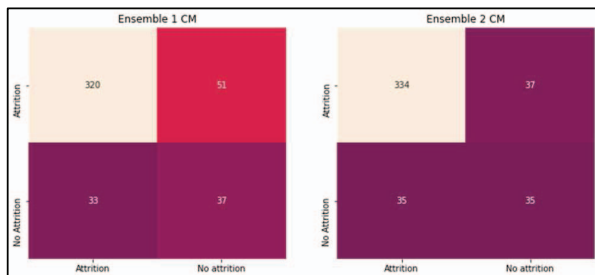


Fig. 6- Prediction result for model ensemble model

VI. DISCUSSIONS

After constructing multiple models, it was observed that three of them had superior performance in forecasting staff attrition.

The Random Forest model, when trained on a subset of selected features, achieved an overall accuracy of 84% and an attrition accuracy of 28.5%.

The implementation of Naive Bayes using all available features. The accuracy rate of 75% and the attrition accuracy rate of 65.7% were observed in the study.

Ensemble 2, which is a fusion of three models incorporating all features, achieved an accuracy rate of 83.7%. Additionally, it demonstrated a 50% attrition accuracy.

It has been noticed that the Naive Bayes model, which is considered the simplest model, demonstrates considerable utility. Although the overall accuracy of the model is relatively lackluster at 75%, its ability to anticipate attrition is highly valuable.

TABLE III. Evaluation result analysis for all models

Model	Evaluation result (Selected Feature)	Evaluation result (New Selected Feature)
	Accuracy	Accuracy
RF	86.39455782312925%	81.40589569160997%
KNN	85.26077097505669%	84.58049886621315%
NB	75.05668934240363%	64.39909297052154%

Model	Evaluation result (With grid search and selected features)		Evaluation result (With grid search and new selected features)	
	Accuracy	F1Score	Accuracy	F1Score
RF	86.16780045351474%	27.587713109657507%	84.12698412698413%	37.98424023994148%
KN	82.76643990929705%	28.97508846288358%	83.21995464852607%	41.10188682558402%

VII. IMPLICATIONS

The analysis conducted in this study provides valuable insights for managers seeking to address attrition and enhance staff retention within the organization.

One should prioritize the task of attrition prediction by implementing models such as Naive Bayes in order to identify prospective cases of attrition through the use of predictive analytics. This enables the organisation to proactively implement strategies aimed at retaining valuable employees prior to their departure.

Allocate resources towards conducting more research to enhance the process of feature selection, with the aim of uncovering a broader range of pertinent and influential elements associated with attrition. This will lead to the development of more accurate prediction models.

Place emphasis on enhancing employee engagement and happiness through the provision of growth prospects, recognition, and demanding tasks. This is particularly important in relation to criteria such as tenure under the same management, duration since the last promotion, and length of service within the organisation.

Since research has linked excessive overtime to employee turnover, workload allocation, and overtime reduction are crucial. To reduce burnout and attrition, introduce work-life balance programmes and flexible scheduling.

Salespeople are more likely to leave, thus they need lots of help. Use customised methods to boost work satisfaction and retention in this demographic.

Establish a diverse and inclusive workplace to promote inclusivity and reduce employee turnover. Encourage open communication, skill development, and mutual respect and admiration.

Periodic and constructive performance assessments provide feedback and help employees develop professionally. Establish fairness and transparency methods to enhance justice and openness in the assessment process. Additionally, employees need clear career development paths.

Exit interviews and comments from departing employees might help identify employee attrition reasons. Use this information to identify issues and make changes to address employee complaints.

VIII. CONCLUSION

The phenomenon of employee churn is a substantial obstacle for organisations on a global scale. This study assesses a range of machine learning techniques, such as Random Forest, K-Nearest Neighbours (K-NN), and Naive Bayes, to make predictions about attrition. The findings suggest that the Naive Bayes model, when utilising all features, is the preferred choice for predicting attrition, while not achieving the maximum overall accuracy.

The examination of the plot analysis uncovers a significant correlation between attrition and several parameters, such as the duration of employment under the same manager, the length of time since the last promotion, the tenure at the organisation, and gender. Sales professionals are likewise prone to experiencing attrition.

These insights offer significant value in comprehending the factors that contribute to attrition inside the organisation. By placing emphasis on these variables, human resources departments can effectively execute specific tactics aimed at enhancing employee retention and cultivating a more steadfast and involved workforce.

In forthcoming times, it is plausible for organisations to delve into more advanced predictive models, such as deep learning and neural networks, in order to apprehend intricate associations between features and enhance attrition predictions. The application of natural language processing in the analysis of employee sentiment can yield more

profound insights into the variables contributing to attrition and facilitate improved selection of features for predictive models. The use of longitudinal studies enables organisations to systematically track attrition patterns over an extended period, thereby providing valuable insights into the efficacy of retention initiatives. The utilisation of employee experience analytics can yield significant information for the purpose of optimising tactics aimed at employee retention.

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